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Task 1:

```
#include<iostream>

using namespace std;

struct symbol{
    string name;
    string type;
    string value;
    string memorySize;
};

int main(){
    symbol table[10];

    int n;

    cout<<"How many variable you want to enter"<<endl;
    cin>>n;

    for(int i=0;i<n;i++){
        cout<<"Enter data type"<<endl;
        cin>>table[i].type;

        cout<<"Enter Variable name"<<endl;
        cin>>table[i].name;

        cout<<"Enter value"<<endl;
        cin>>table[i].value;
    }

    cout<<"-----Symbol table-----\n"<<endl;
```

```

cout<<"\n\ttype \tname \tvalue \tsize\n";
for(int i=0;i<n;i++)
{
    cout<<table[i].name<<"\t";
    cout<<table[i].type<<"\t";
    cout<<table[i].value<<"\t";
    for(int i=0;i<n;i++)
    {
        if(table[i].type=="int")
        {
            table[i].memorySize = "4 Bytes";
        }
        if(table[i].type=="float")
        {
            table[i].memorySize = "4 Bytes";
        }
        if(table[i].type=="string")
        {
            table[i].memorySize = "4 Bytes";
        }
    }
    cout<<table[i].memorySize<<"\t";
}

return 0;
}

```

```

How many variable you want to enter
2
Enter data type
int
Enter Variable name
a
Enter value
3
Enter data type
float
Enter Variable name
b
Enter value
3.4
-----Symbol table-----
type    name    value    size
a       int      3        4 Bytes b    float   3.4     4 Bytes
-----
Process exited after 20.89 seconds with return value 0
Press any key to continue . . .

```

Task 2:

```
#include <iostream>
```

```
using namespace std;
```

```
struct Symbol {
```

```
    string name;
```

```
    string type;
```

```
    int address;
```

```
};
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter number of identifiers: ";
```

```
    cin >> n;
```

```
    Symbol table[10];
```

```
for (int i = 0; i < n; i++) {  
    cout << "\nEnter details for identifier " << i + 1 << endl;  
    cout << "Variable Name: ";  
    cin >> table[i].name;  
    cout << "Type: ";  
    cin >> table[i].type;  
    cout << "Address: ";  
    cin >> table[i].address;  
}
```

```
string searchName;  
cout << "\nEnter identifier variable name to search: ";  
cin >> searchName;
```

```
bool found = false;
```

```
for (int i = 0; i < n; i++) {  
    if (table[i].name == searchName) {  
        cout << "\nIdentifier Found!\n";  
        cout << "Name: " << table[i].name << endl;  
        cout << "Type: " << table[i].type << endl;  
        cout << "Address: " << table[i].address << endl;  
        found = true;  
        break;  
    }  
}
```

```

    if(!found) {
        cout << "\nIdentifier not found.\n";
    }

    return 0;
}

```

```

Enter number of identifiers: 2
Enter details for identifier 1
Variable Name: a
Type: int
Address: 1234
Enter details for identifier 2
Variable Name: b
Type: float
Address: 1236

Enter identifier variable name to search: a

Identifier Found!
Name: a
Type: int
Address: 1234
-----
Process exited after 36.65 seconds with return value 0
Press any key to continue . . .

```

Task 3:

```

#include <iostream>

using namespace std;

```

```

struct Symbol {
    string name;
    string type;
    int address;
};

```

```

int main() {
    Symbol table[100];

    int n;

    int count = 0;

```

```
cout << "Enter number of identifiers to insert: ";
cin >> n;

for (int i = 0; i < n; i++) {
    string name, type;
    int address;
    bool duplicate = false;

    cout << "\nEnter identifier name: ";
    cin >> name;

    // Check for duplicate
    for (int j = 0; j < count; j++) {
        if (table[j].name == name) {
            duplicate = true;
            break;
        }
    }

    if (duplicate) {
        cout << "Error: Duplicate identifier! Insertion not allowed.\n";
    } else {
        cout << "Enter type: ";
        cin >> type;
        cout << "Enter address: ";
        cin >> address;

        table[count].name = name;
```

```
    table[count].type = type;

    table[count].address = address;

    count++;

    cout << "Identifier inserted successfully.\n";
}
}

// Display final symbol table
cout << "\nFinal Symbol Table (Unique Identifiers Only):\n";
for (int i = 0; i < count; i++) {
    cout << "\nName: " << table[i].name;
    cout << "\nType: " << table[i].type;
    cout << "\nAddress: " << table[i].address << endl;
}

return 0;
}
```

```
Enter number of identifiers to insert: 2
Enter identifier name: a
Enter type: int
Enter address: 236
Identifier inserted successfully.
Enter identifier name: c
Enter type: float
Enter address: 786545
Identifier inserted successfully.

Final Symbol Table (Unique Identifiers Only):
Name: a
Type: int
Address: 236
Name: c
Type: float
Address: 786545
-----
Process exited after 26.41 seconds with return value 0
Press any key to continue . . .
```